THE AFFINE ETALE SITE OF A DIRECTED LIMIT OF SCHEMES IS THE COLIMIT OF THE AFFINE ETALE SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both cases, with the reduction from the general case to the quasi-compact one carried out rather than quoted.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Affine etale site of a scheme. Topologies, the site whose objects are affine schemes etale over the scheme, with the etale topology. Supplies the two sides of the asserted equality of sites.
- Colimit of sites along a directed inverse system. Sites, the construction of the colimit of a system of sites and the criterion for recognising an isomorphism of sites. Supplies the sense in which the displayed equality is asserted and the criterion used in the proof.
- Affine transition morphisms and base change along a limit of schemes. Same chapter, the description of quasi-compact and quasi-separated objects of the affine etale site of a limit as base changes of objects over some index. Supplies the general case of the proof, in which an object of the limit site is descended to some stage.

Source reference. Original lemma 04HR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7340–7340. Navigation pointer invoked at proof line 7340; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09YL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7345–7345. Navigation pointer invoked at proof

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line 7345; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01ZM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7357–7357. Navigation pointer invoked at proof line 7357; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01Z6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7360–7360. Navigation pointer invoked at proof line 7360; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07RP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7360–7360. Navigation pointer invoked at proof line 7360; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07RR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7366–7366. Navigation pointer invoked at proof line 7366; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EY1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7373–7373. Navigation pointer invoked at proof line 7373; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let I be a directed set. Let $(X_i, f_{i'i})$ be an inverse system of schemes over I with affine transition morphisms. Let $X = \lim_{i \in I} X_i$. With notation as in Topologies, Lemma 04HR we have*

$$X_{affine,étale} = \mathrm{colim}(X_i)_{affine,étale}$$

as sites in the sense of Sites, Lemma 09YL.

Proof. Let us first prove this when X and X_i are quasi-compact and quasi-separated for all i (as this is true in all cases of interest). In this case any object of $X_{affine,étale}$, resp. $(X_i)_{affine,étale}$ is of finite presentation over X. Moreover, the category of schemes of finite presentation over X is the colimit of the categories of schemes of finite presentation over X_i, see Limits, Lemma 01ZM. The same holds for the subcategories of affine objects étale over X by Limits, Lemmas 01Z6 and 07RP. Finally, if $\{U^j \to U\}$ is a covering of $X_{affine,étale}$ and if $U_i^j \to U_i$ is morphism of affine schemes étale over X_i whose base change to X is $U^j \to U$, then we see that the base change of $\{U_i^j \to U_i\}$ to some $X_{i'}$ is a covering for i' large enough, see Limits, Lemma 07RR.

In the general case, let U be an object of $X_{affine,étale}$. Then $U \to X$ is étale and separated (as U is separated) but in general not quasi-compact. Still, $U \to X$ is locally of finite presentation and hence by Limits, Lemma 0EY1 there exists an i, a quasi-compact and quasi-separated scheme U_i, and a morphism $U_i \to X_i$ which is locally of finite presentation whose base change to X is $U \to X$. Then

$U = \lim_{i' \geq i} U_{i'}$ where $U_{i'} = U_i \times_{X_i} X_{i'}$. After increasing i we may assume U_i is affine, see Limits, Lemma 01Z6. To check that $U_i \to X_i$ is étale for i sufficiently large, choose a finite affine open covering $U_i = U_{i,1} \cup \ldots \cup U_{i,m}$ such that $U_{i,j} \to U_i \to X_i$ maps into an affine open $W_{i,j} \subset X_i$. Then we can apply Limits, Lemma 07RP to see that $U_{i,j} \to W_{i,j}$ is étale after possibly increasing i. In this way we see that the functor $\mathrm{colim}(X_i)_{affine,étale} \to X_{affine,étale}$ is essentially surjective. Fully faithfulness follows directly from the already used Limits, Lemma 0EY1. The statement on coverings is proved in exactly the same manner as done in the first paragraph of the proof. □